



Abylay Bissequenov

Astrophysicist

📍 Suzhou, China ✉ abylay.bissekenov22@student.xjtlu.edu.cn

🌐 abissekenov@github.io 🔗 Abylay Bissequenov 🌐 abissekenov

Education

PhD	Xi'an Jiaotong-Liverpool University , Physics	Suzhou, China
	- Expected graduation at the end of 2026. - Upon delays, the CV would be updated	Mar 2024 – Dec 2026
MS	Nazarbayev University , Physics	Astana, Kazakhstan
	- GPA: 3.5/4.00 - Graduated with honors	Aug 2021 – June 2023
BS	K.Zhubanov Aktobe Regional State University , Physics	Aktobe, Kazakhstan
	- GPA: 3.64/4.00 - Graduated with honors	Sept 2016 – July 2020

Experience

Xi'an Jiaotong-Liverpool University , Research Assistant	Suzhou, China
	Mar 2024 – present 2 years 4 months
Xi'an Jiaotong-Liverpool University , Teaching Assistant	Suzhou, China
Over 500 hours of TAs experience	Mar 2024 – June 2026 2 years 4 months
Energetic Cosmos Laboratory , Research Assistant	Astana, Kazakhstan
Worked under supervision of Bekdaulet Shukirgaliyev and Ernazar Abdikamalov on applications of ML techniques on N-body simulations.	June 2022 – Dec 2024 2 years 7 months

Publications

Formation and Disruption of Wide Binaries in Star Clusters Revealed by N-body Simulations	Apr 2026
Abylay Bissequenov, Xiaoying Pang, Rainer Spurzem, Bekdaulet Shukirgaliyev, Mukhagali Kalambay, Peter Berczik	
10.3847/2041-8213/ae5819 (The Astrophysical Journal Letters)	
Evolution of star clusters with initial bulk rotation via N-body simulations	July 2025
Abylay Bissequenov, Xiaoying Pang, Kamlah Albrecht, M. B. N. Kouwenhoven, Rainer Spurzem, Bekdaulet Shukirgaliyev, Mirek Giersz, Abbas Askar, Peter Berczik	
10.1051/0004-6361/202554093 (Astronomy and Astrophysics)	
Cluster membership analysis with supervised learning and N-body simulations	Sept 2024
Abylay Bissequenov, Mukhagali Kalambay, Ernazar Abdikamalov, Xiaoying Pang, Peter Berczik, Bekdaulet Shukirgaliyev	
10.1234/icml.2022.9012 (Astronomy and Astrophysics)	

Conferences, Talks and Visits

- European Astronomical Society Annual Meeting 2026; Lausanne, Switzerland; 2026/06/29-2026/07/04 - K-Poster Presentation
- 1st Silk Road Conference; Tashkent, Uzbekistan; 2026/05/18-2026/05/22 - Contrabuted Presentation
- Structural, Formation and Evolution of Clusters 2026 Annual Symposium; Nanjing, China; 2026/04/22-2026/04/26 - Contrabuted Presentation
- 2nd XJTU-XJTLU-UoL Doctoral Wisdom Convergence Camp; Xi'an, China; 2026/04/17-2026/04/19 - Oral Presentation
- 2026 XJTLU Postgraduate Research Symposium; Suzhou, China; 2026/03/19-2026/03/20 - Oral and Poster Presentation
- Informal Presentation at Fesenkov Astrophysical Institute; 2025/10/08 - Oral Presentation
- Informal Presentation at Al-Farabi Kazakh National University; 2025/10/07 - Oral Presentation
- Informal Presentation at Nazarbayev University; 2025/09/29 - Oral Presentation
- Galaxy Coffee Seminar at Max Plank Institute of Astronomy; 2025/07/29 - Oral Presentation
- Researcher visit to University of Heidelberg; Heidelberg, Germany; 2025/06/29-2025/08/04
- Galaxy Evolution Seminar of Eva K. Grebel group in University of Heidelberg; 2025/07/16 - Oral Presentation
- IAU Symposium 398/MODEST-25; Seoul, South Korea; 2025/06/15-2025/06/20 - Contrabuted Presentation
- 2025 XJTLU Postgraduate Research Symposium; Suzhou, China; 2025/03/19-2025/03/21 - Oral and Poster Presentation
- Structural, Formation and Evolution of Clusters 2024 Annual Symposium; Hangzhou, China; 2025/04/26-2025/04/29 - Contrabuted Presentation
- Academic mobility to Gdansk University; Gdansk, Poland; 2017/10/01-2018/01/25

Awards

- Best Oral Presentation Award - 2nd XJTU-XJTLU-UoL Doctoral Wisdom Convergence Camp (2026-04)
- Best Poster Presentation Award - XJTLU Postgraduate Research Symposium (2026-03)
- Excellent Oral and Poster Presentation Awards - XJTLU Postgraduate Research Symposium (2025-03)
- Deans List - Nazarbayev University (2023-09)

Skills

Languages: Python, C/C++, HTML/CSS, Fortran

ML Frameworks: PyTorch, TensorFlow

Research Areas: Astrophysics, Physics, Data Science, High-Performance Computing